Sec. VI-E: hand-calculable bounds

numbering continues from (105)

Eq. (105) bounds the shifted critical moment by the two channel maxima, and (93) supplies the constants that multiply them. What remains is to evaluate $\rho_{\varphi,\max}$ and $\rho_{GE,\max}$ themselves. Both are collected here rather than at the point where each channel was introduced, so that the arithmetic appears once.

Nothing below requires a run. The weight comes from a scale, the pivot arm and z_{CoM} from the box of Sec. VI-D, the inertia from CAD, the tilt cap and the ramp rates from the excitation protocol of Sec. V-B, and the ground-effect slope from table 3. The bound is therefore a statement made before the campaign, not a description of it.

The window, which only the coupling channel needs. The nominal excursion of (20) is $\varphi_{\text{end}} = \big(\dot M/W z_{CoM} C_2\big)\Lambda(x)$, and every term of Λ is positive, so $\Lambda(x) \geq x^3/6$. Imposing the tilt cap $\varphi_{\text{end}} \leq \varphi_{\max}$ therefore inverts in closed form, and with $J_P = W z_{CoM}/C_2^2$ the window length and the moment increment across it follow without solving for x:

$$\tau_{\text{end}} \leq \left(\frac{6\varphi_{\max} J_P}{\dot M}\right)^{1/3}, \qquad \Delta M_{\text{win}} = \dot M \tau_{\text{end}} \leq \big(6\varphi_{\max} J_P \dot M^2\big)^{1/3}. \tag{106}$$

The inertia enters at its upper end, $J_P \leq J_{CoM}^{\text{CAD}} + m\big(z_{CoM}^2 + \big(l_p + p_{\text{off}}\big)^2\big)$, which is the unfavourable side here: (82) discards that CAD term, but retaining it is what makes (106) a bound rather than an estimate.

The gravity channel. By (85) the remainder is $\rho_\varphi = \frac{1}{2}G''(\varphi^*)\varphi^2$ with $G''(\varphi^*) = W\big(l_p + p_{\text{off}}\big)\cos\varphi^* - W z_{CoM}\sin\varphi^*$, which decreases on $[0, \varphi_{\max}]$ and is therefore largest at $\varphi^* = 0$. Hence

$$\rho_{\varphi,\max} \;\leq\; \tfrac{1}{2}\, W\big(l_p + p_{\text{off}}\big)\,\varphi_{\max}^2, \tag{107}$$

evaluated at the ballasted weight and at the widest arm the offset box admits. It does not depend on the ramp rate: the tilt cap alone fixes it.

The coupling channel. Term (iv) of (80) is $\rho_{GE}(\tau,\varphi) = \beta_M \dot M \tau\varphi$, so its maximum over the window is the product of the moment increment and the excursion, both already bounded:

$$\rho_{GE,\max} = |\beta_M|\, \Delta M_{\text{win}}\, \varphi_{\text{end}} \;\leq\; |\beta_M|\, \varphi_{\max}\big(6\varphi_{\max} J_P \dot M^2\big)^{1/3}. \tag{108}$$

This is the only place the ramp rate enters, and it enters as $\dot M^{2/3}$.

The bound, in six lines. Substituting (107) and (108) into (105) and dividing by the unloaded weight gives the equivalent offset error. No hyperbolic function appears, C_2 never enters, and x is never solved for — $\frac{1}{7}$ and $\frac{1}{5}$ in (93) are suprema over *all* x:

$$\boxed{\big|\Delta M_{\text{crit},e}\big| \leq \frac{\rho_{\varphi,\max}}{7} + \frac{\rho_{GE,\max}}{5}, \qquad \big|\Delta\lambda_{\text{off}}\big| \leq \frac{1}{W}\left[\frac{\rho_{\varphi,\max}}{7} + \frac{\rho_{GE,\max}}{5}\right].} \tag{109}$$

Over the admissible box, with $W = 31.59$ N, $z_{CoM} = 0.30$ m, $J_{CoM}^{\text{CAD}} = 0.0537\ \text{kg}\,\text{m}^2$, $\varphi_{\max} = 10°$ and the arms and slopes above:

axis	$\dot M$ [N m/s]	J_P [kg m^2]	ΔM_{win} [N m]	$\rho_{\varphi,\max}$ [mN m]	$\rho_{GE,\max}$ [mN m]	$\lvert\Delta\lambda_{\text{off}}\rvert$ [mm]
roll	0.10	0.4260	0.165	76.98	0.99	0.372
	0.45	0.4260	0.449	76.98	2.70	0.384
	1.20	0.4260	0.863	76.98	5.19	0.400
pitch	0.10	0.3979	0.161	62.55	0.72	0.302
	0.45	0.3979	0.439	62.55	1.97	0.310
	1.20	0.3979	0.843	62.55	3.79	0.322

The same row, by hand. Roll at $\dot{M} = 1.20$ N m/s, every intermediate carried so the row can be checked on a calculator:

	quantity	value
	$\varphi_{\max} = 10°$	0.174533 rad
	$m = W/g = 31.59/9.81$	3.220183 kg
	$z_{CoM}^2 + (l_p + p_{\text{off}})^2 = 0.09 + 0.0256$	0.115600 m^2
	$m(z_{CoM}^2 + (l_p + p_{\text{off}})^2)$	0.372253 kg m^2
(1)	$J_P \leq 0.0537 + 0.372253$	0.425953 kg m^2
	$6\varphi_{\max} J_P \dot{M}^2 = 6(0.174533)(0.425953)(1.44)$	0.642322
(2)	$\Delta M_{\text{win}} \leq (0.642322)^{1/3}$	0.862815 N m
	$\varphi_{\max}^2$	0.030462
(3)	$\rho_{\varphi,\max} = \frac{1}{2}(31.59)(0.160)(0.030462)$	0.076983 N m
(4)	$\rho_{GE,\max} = (0.03446)(0.862815)(0.174533)$	0.005189 N m
	$\rho_{\varphi,\max}/7$	10.998 mN m
	$\rho_{GE,\max}/5$	1.038 mN m
(5)	$\lvert \Delta M_{\text{crit},e} \rvert \leq$ sum	12.035 mN m
(6)	$\div W = 30.08$ N	0.400 mm

Note which weight goes where: $W = 31.59$ N, the ballasted value, enters (3) because a heavier vehicle makes the gravity remainder larger, while $W = 30.08$ N, the unloaded value, enters (6) because a lighter vehicle turns a given moment error into a larger offset. Each is taken at whichever end of the range is unfavourable for the quantity it appears in.

The worst admissible case is 0.400 mm for roll and 0.322 mm for pitch, of which the gravity channel supplies 91 and 92%. Two readings follow. The inertia, the one input that is not directly measured, reaches the result only through $\rho_{GE,\max}$ and only as a cube root: halving J_P moves the bound by -1.8% and doubling it by $+2.2\%$, so a coarse CAD figure suffices. And precision is bought at the tilt cap rather than at the ramp: (107) is quadratic in $\varphi_{\max}$ while (108) is only $\dot{M}^{2/3}$, so a twelve-fold reduction in ramp rate moves the bound by 7% whereas halving the cap quarters it. The 10° used here was chosen for onset visibility on the lightly loaded configurations; at 5° the worst case would fall to 0.105 mm for roll and 0.084 mm for pitch — a little short of a factor of four, because $\rho_{GE,\max}$ carries only $\varphi_{\max}^{4/3}$ — at the price of a 21% shorter fitting window and a rate at the cap that drops from 14.8 to 2.9 rad/s on the slowest ramp.

The angular rate error. Eq. (90) bounds the rate deviation itself, and unlike (109) it needs the window. The same cubic inversion gives $x \leq \bar{x} = \left(6\varphi_{\max} W z_{CoM} C_2/\dot{M}\right)^{1/3}$, and C_2 has a geometric ceiling in which the mass cancels,

$$C_2^2 = \frac{W z_{CoM}}{J_P} \leq \frac{W z_{CoM}}{m\left(z_{CoM}^2 + \left(l_p + p_{\text{off}}\right)^2\right)} = \frac{g\, z_{CoM}}{z_{CoM}^2 + \left(l_p + p_{\text{off}}\right)^2}, \tag{110}$$

giving 5.05 s^{-1} for roll and 5.25 for pitch. Note that here the inertia is taken at its *lower* end, $J_P = m(z_{CoM}^2 + (l_p + p_{\text{off}})^2)$, the opposite of (106): $J_P C_2 = \sqrt{J_P W z_{CoM}}$ sits in the denominator of (90), and a smaller J_P both enlarges that quotient and raises C_2, hence $\bar{x}$. Continuing the roll row above:

	quantity	value
	$J_P = m(z_{CoM}^2 + (l_p + p_{\text{off}})^2)$, lower end	0.372253 kg m^2
	$W z_{CoM} = (31.59)(0.30)$	9.477000 N m
(7)	$C_2 = \sqrt{W z_{CoM}/J_P}$, and (110) agrees	5.045639 s^{-1}
	$6\varphi_{\max} W z_{CoM} C_2/\dot{M} = 6(0.174533)(9.477)(5.045639)/1.20$	41.728663
(8)	$\bar{x} = (41.728663)^{1/3}$	3.468525
	$\sinh \bar{x}$	16.029105
	$J_P C_2 = \sqrt{J_P W z_{CoM}}$	1.878255
(9)	$\lvert e_\omega(\tau_{\text{end}}) \rvert \leq \bar{\rho} \sinh \bar{x}/(J_P C_2)$, $\bar{\rho} = 0.012035$	0.102707 rad/s
	$C_1 = \dot{M}/W z_{CoM}$	0.126622
	$\cosh \bar{x} - 1$	15.060268
(10)	$\omega_{\text{nom,end}} = C_1(\cosh \bar{x} - 1)$	1.906967 rad/s
(11)	relative, (9) $\div$ (10)	5.39%

The $\bar{\rho}$ in (9) is the same 12.035 mN m that line (5) produced; (93) bounds the average, and (90) then propagates it through the kernel. Across the ramp range the roll axis gives

$\dot{M}$ [N m/s]	$\bar{x}$	$\lvert e_\omega \rvert$ [rad/s]	$\omega_{\text{nom,end}}$ [rad/s]	relative
0.10	7.94	8.378	14.815	56.6%
0.45	4.81	0.377	2.866	13.2%
1.20	3.47	0.103	1.907	5.4%

The slow ramp is where this bound stops being useful, and the reason is visible in the arithmetic: $\bar{x}$ grows as $\dot{M}^{-1/3}$ but enters through sinh, so the cubic shortcut $\Lambda(x) \geq x^3/6$ that makes (106) closed-form is exponentially lossy once $\bar{x}$ passes about six. That degradation does not touch (109), which carries no x at all — which is why the threshold bound, and not the rate bound, is what the offset budget inherits.

What (109) covers. The displacement of the identified onset caused by the two forcing terms the estimator cannot absorb, and nothing else. The baseline error of (104), the departure of the applied moment from a linear ramp, and the ground-contact geometry all lie outside it. Those are measured rather than bounded a priori, and are reported with the realised budget in Sec. VIII-E, where the same runs give an actual propagated error some fifty times inside (109).